An analysis of the impact of
AI demand on DRAM Prices

In recent years, one thing has become clear: artificial intelligence (AI) is shaping the future. From simple art generation to even predictive medical evaluations, it seems that each and every industry is incorporating AI into their workings. However, one point that consistently dominates AI debate is its impact on the prices for memory, specifically DRAM. It is argued that an over-reliance on AI is inflating the prices of memory, resulting in higher prices across the globe for technologies such as video game consoles and personal computers (PCs). It is because of this; I have chosen to research the impact of AI demand on the prices of DRAM using a simple ordinary least squares (OLS) regression to monitor AI's impact on DRAM markets.

My methodology involved running two simple OLS regressions: One to assess the impact of AI demand directly on DRAM prices, and another one to assess the impact of HBM shares and DRAM per PC (which would also be influenced by AI demand) on DRAM prices. Both regressions used the demand of PCs as a control variable to allow for regression results to speak only for our independent variables. Regression one followed the form: DRAM Price = $\beta_0 + \beta_1(PC\ Demand) + \beta_2(AI\ Demand) + \varepsilon$ and regression two followed the form: DRAM Price = $\beta_0 + \beta_1(PC\ Demand) + \beta_2(HBM\ Share) + \beta_3(DRAM\ per\ PC) + \varepsilon$. To measure DRAM Prices, I looked at historical DRAM prices per quarter for a gigabyte (GB) in USD. PC Demand was measured by tracking global shipments of PCs in millions per quarter with figures being sourced from various Gartner and IDC reports. AI Demand was measured by tracking the combined data centre revenues of Nvidia, Intel and AMD per quarter. HBM share and DRAM per PC was measured by taking the percentage share of HBM in total DRAM bit capacity and the DRAM per PC measured in gigabytes from Gartner reports. Each variable for Regression 1 has per quarter values starting from the second quarter of 2021 to the fourth quarter of 2025. This allows me to have sufficient data to run a meaningful regression and analyse the direct impacts of the AI boom without other significant macroeconomic events influencing the data. Regression 2, due to insufficient data, only uses per annum recordings. This has consequences for the regression which will be discussed later. For each β_x analysed, it will be tested against the hypothesis H_0: $\beta_x = 0$, H_1: $\beta_x \neq 0$ using the appropriate t-values

Using a Python script to run and analyse the regression, my findings were that contrary to popular belief AI demand seemed to have no overall negative effect on DRAM prices. However, indirect AI demand measured through HBM shares and DRAM per PC may have an influence on DRAM prices. Regression 1 had the finding $\beta_2 = -0.0012$. This means that, ceteris paribus, for every billion U.S. dollars of revenue generated by data centres, the price of DRAM had fallen by 0.0012 USD per GB, with a t value of -0.256, indicating that this result is not statistically significant at the 5 percent significance level, as it does not reach the critical value of -1.96. In fact, this t-value is so low that it expresses zero direct linear relationship between quarterly data centre revenue and quarterly DRAM prices. Regression 2 finds that $\beta_2 = 0.6555$ and $\beta_3 = -0.7125$. This means that, ceteris paribus, for ever percent of HBM share of total DRAM bit capacity, DRAM prices are expected to increase by 0.6555 USD per GB and, ceteris paribus, for every gigabyte of DRAM per PC, DRAM prices are expected to decrease by 0.7125 USD per GB. Both would have statistically significant t-values at the 5% significance level, with values of 2.389 and -2.994 respectively, however due to the small sample size and low degrees of freedom, the critical values for this regression would be 3.182 or -3.182 – neither variable meets these criteria and thus neither are statistically significant.

Whilst I had attempted to ensure each regression could act as the best linear unbiased estimator for DRAM prices, some issues had been faced. Due to utilising global data, I could not control for the impact of inflation (which would have different values per country per annum) nor could I control for the growth and health of the tech industry in each country. This means that there could be significant omitted variable bias which could impact the results found. Furthermore, the utilisation of per annum data for regression 2 means that the sample size for this regression is extremely small, with only 7 observations. This means that the results found for this regression are most likely to be inaccurate, which is reflected by the extraordinarily high condition number seen in figure A. If I were to attempt this project again, I would likely focus on one country or region such that I could get more accurate data and control for more variables such as the ones mentioned above. I would also attempt to find a better proxy for AI demand rather than data centre revenue as this proxy would also include non-AI data centre revenue.

From the regressions we can't exactly make a definite answer. The first regression seems to indicate that AI demand does not have any statistical impact on DRAM prices whatsoever. However, the second regression shines a light onto why this might be the case. We saw that HBM share of the total DRAM bit capacity has quite a significant positive effect on the price of DRAM, because HBM is primarily used in AI hardware, it is a good indicator of AI demand. We also saw that DRAM per PC had quite a significant negative effect on the price of DRAM, this is because DRAM per PC reflects consumer trends as well as AI trends. If DRAM prices were to fall, then we would see consumers increase their DRAM per PC – it could be that we have found a reverse correlation. Moreover, this upwards effect from HBM shares with the downwards effect of DRAM per PC may explain the overall neutral effect of direct AI demand.

To conclude, whilst we can see some very interesting results when analysing this problem from a statistical lens, a regression alone (and quite a flawed regression model) cannot accurately explore the impact of AI demand on DRAM prices. Instead, a mixture of empirical analysis and economic theory should be used. AI is shaping the future, and it serves as an excellent case study for economists much better than myself to review the impacts of a technology boom on global markets as well as many other interesting questions.

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DATA:
DRAM Prices: https://dam.stanford.edu/memory-prices.html

PC Shipments: https://www.statista.com/statistics/534779/worldwide-pc-shipments-perquarter/

Data centre segment revenue: https://www.statista.com/statistics/1425087/data-centersegment-revenue-nvidia-amd-intel/

HBM shares and DRAM per PC sourced from Gartner Semiconductor Tracker.

Figure A:

An analysis of the impact of
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21]:
```
print(reg1.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              DRAMPrice   R-squared:                       0.684
Model:                            OLS   Adj. R-squared:                  0.644
Method:                 Least Squares   F-statistic:                     17.28
Date:                Fri, 10 Jul 2026   Prob (F-statistic):           0.000100
Time:                        11:34:43   Log-Likelihood:                -7.6752
No. Observations:                  19   AIC:                             21.35
Df Residuals:                      16   BIC:                             24.18
Df Model:                           2
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept            -1.8932      0.750     -2.525      0.023      -3.483      -0.304
Shipments             0.0572      0.010      5.750      0.000       0.036       0.078
DataCenterRevenue    -0.0012      0.005     -0.256      0.801      -0.011       0.009
==============================================================================
Omnibus:                        0.220   Durbin-Watson:                   1.531
Prob(Omnibus):                  0.896   Jarque-Bera (JB):                0.270
Skew:                           0.210   Prob(JB):                        0.874
Kurtosis:                       2.594   Cond. No.                         638.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

22]:
```
print(reg2.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:              DRAMPrice   R-squared:                       0.885
Model:                            OLS   Adj. R-squared:                  0.770
Method:                 Least Squares   F-statistic:                     7.712
Date:                Fri, 10 Jul 2026   Prob (F-statistic):             0.0637
Time:                        11:35:24   Log-Likelihood:                -1.5305
No. Observations:                   7   AIC:                             11.06
Df Residuals:                       3   BIC:                             10.84
Df Model:                           3
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept      7.1112      2.714      2.620      0.079      -1.527      15.749
HBM_Share      0.6555      0.274      2.389      0.097      -0.218       1.529
DRAM_PerPC    -0.7125      0.238     -2.994      0.058      -1.470       0.045
Shipments      0.0128      0.006      2.058      0.132      -0.007       0.033
==============================================================================
Omnibus:                          nan   Durbin-Watson:                   2.773
Prob(Omnibus):                    nan   Jarque-Bera (JB):                0.859
Skew:                          -0.852   Prob(JB):                        0.651
Kurtosis:                       2.793   Cond. No.                     4.41e+03
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 4.41e+03. This might indicate that there are
strong multicollinearity or other numerical problems.
```